

Optimising Data Analytics to Enhance Postgraduate Student Academic Achievement: A Systematic Review

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Abstract: This systematic review investigated how Higher Education Institutions (HEIs) optimise data analytics in postgraduate programmes to enhance student achievement. Existing research explores the theoretical benefits of data analytics but lacks practical guidance on strategies to effectively implement and utilise data analytics for student success. As such, this review aimed to identify data analytics approaches used by HEIs and explore challenges and best practices in their application. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were followed. Five databases were searched. Studies that examined data analytics in HEIs postgraduate programmes and their impact on student learning were included. Studies that were solely theoretical or in non-postgraduate settings were excluded. Twenty-six studies were included. Quality assessment using the Critical Appraisal Skills Programme (CASP) Checklist was employed. The review identified various data analytics approaches including descriptive, predictive, and prescriptive analytics, among others. These approaches can improve foundational skills, create supportive learning environments, and optimise teaching strategies. However, limitations (standardised tests, data integration) and privacy concerns were acknowledged. Recommendations include developing a comprehensive evaluation system, equipping educators with the skills to utilise diverse analytics to enhance student achievement, fostering open communication about data use, and cultivating a data-literate student body. While diverse approaches were explored, the review's lack of specific contextual details may limit the generalisability of findings. To mitigate this, the review categorised techniques and provided references for further exploration.



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Keywords: data analytics; higher education institutions; learning analytics; learning experiences; postgraduate education; systematic review

1. Introduction

Postgraduate education, encompassing programmes designed for students with bachelor's degrees seeking advanced qualifications such as postgraduate diplomas, master's degrees, or doctoral studies, provides a distinctive context for harnessing data analytics to enhance academic success. [1,2]. Academic achievement refers to the progress of acquiring educational skills, knowledge, and competencies [3]. It is typically measured through various assessment methods and reflects a student's ability to meet the programme's learning objectives [4]. As such, academic achievement thrives on a complex interplay of factors relevant to postgraduate studies:

- **Foundational Skills:** Strong foundational literacy, numeracy, and critical thinking skills are crucial for postgraduate success. These skills provide a firm foundation for independent learning and research, which is paramount in postgraduate programs [5].
- **Motivation and Engagement:** Intrinsic motivation, fuelled by a devotion to learning and opportunities for autonomy, is crucial for student engagement across all educational levels. Postgraduate programmes can foster this by offering research

opportunities and independent study options and encouraging student choice in coursework [6,7].

- **Effective Teaching and Learning Strategies:** Effective teaching strategies involve differentiated instruction that caters to diverse learning styles [8,9]. This is particularly important in postgraduate education, where students come from varied backgrounds and experiences. Blended learning approaches that combine face-to-face instruction with online learning platforms can cater to various learning preferences and provide access to specialised resources [10,11].
- **Assessment and Evaluation:** Effective assessment uses various methods to gauge student learning and inform instruction through data-driven decision-making [12]. Formative assessments with clear learning objectives and timely feedback allow students to adjust their learning strategies [13]. Assessments should prioritise learning development over grades to encourage exploration and deep learning [14]. This is particularly important in fostering a growth mindset and a love of research in postgraduate students [12].
- **Supportive Learning Environment:** This requires strong leadership that prioritises student success [15]. Qualified and enthusiastic faculty are crucial [16]. In postgraduate programmes, these educators may be professors, advisors, or research supervisors who guide students and provide mentorship [17]. HEI often provides resources and support services to promote student well-being and create a positive learning climate, which is equally important for postgraduate students navigating coursework, research, and potentially balancing work and personal life [18].

Understanding these diverse factors through data analytics empowers postgraduate programmes to create a data-driven approach [1]. This approach can enhance student learning, personalise the learning experience, and ensure academic achievement [2,12]. Figure 1 depicts the interplay between these factors and student success in postgraduate programmes:

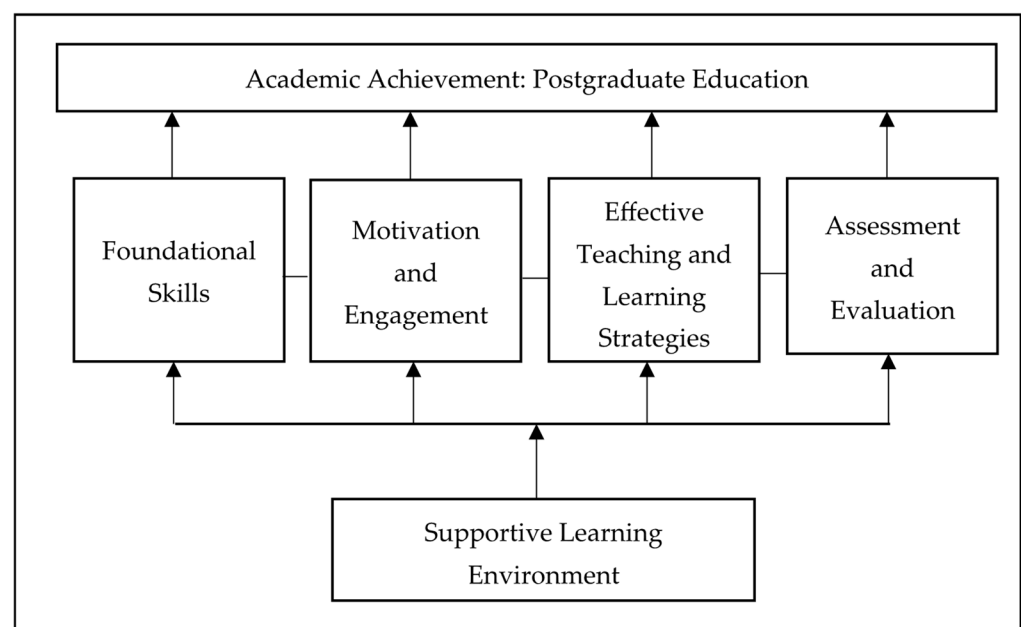


Figure 1. Academic achievement in the context of postgraduate education.

The unique potential of data analytics to enhance postgraduate academic achievement becomes evident when considering the limitations of traditional support methods. For instance, while traditional support methods, like one-size-fits-all study skills workshops [19] and generic time management seminars [20] are common, they lack the precision to address individual student needs. These broad approaches often struggle to identify at-risk

students early [20,21] and offer limited customisation, hindering the personalisation of learning experiences [22]. The absence of personalisation can decrease student engagement and limit academic achievement [23]. Data analytics, on the other hand, offers a promising solution. This methodical process of evaluating and analysing large datasets using statistical and computational techniques holds immense potential to transform postgraduate education [24–27]. Hence, by uncovering meaningful insights and patterns within vast datasets of student information, data analytics can empower educators to:

- Identify trends, correlations, and relationships relevant to postgraduate learning.
- Discern valuable signals from the available data (e.g., student performance, engagement metrics, resource utilisation).
- Make informed decisions regarding programme optimisation, support services, and personalised learning experiences—all grounded in empirical evidence.

Therefore, this ability to harness data-driven insights facilitates the generation of new knowledge about student learning in the postgraduate context. It allows for improved decision-making processes within institutions, fostering innovation in the delivery and support of postgraduate education [28,29]. Educators can effectively employ data analytics to personalise their teaching approaches through face-to-face, online, or blended learning modalities, thereby addressing individual student needs and fostering academic achievement [30].

Integrating various data analytics approaches into postgraduate education offers a deeper understanding of student behaviours and academic performance. These approaches can be categorised as follows:

- **Descriptive Analytics:** This foundational approach summarises and describes past data [31]. Descriptive analytics can be used to understand trends in student enrolment, course completion rates, grades across programs, and demographics of postgraduate students [32]. For example, descriptive analysis might reveal a programme with consistently low completion rates, prompting further investigation into potential causes.
- **Predictive Analytics:** This technique leverages historical data to predict future outcomes [33]. In postgraduate education, predictive analytics can identify students at risk of dropping out, underperforming, or needing additional support. Early identification allows for targeted interventions to improve student success rates. A predictive model might analyse past student data points like prior academic performance, engagement indicators, and course selection to predict students at risk of low grades [34,35].
- **Prescriptive Analytics:** Building upon predictive models, prescriptive analytics goes a step further by recommending specific actions based on the predicted outcomes [36,37]. This allows educators to proactively personalise learning experiences by providing targeted support to at-risk students. For instance, a prescriptive analytics system might recommend additional resources, personalised study plans, or early mentorship for students predicted to struggle in a particular course [38].
- **Text Analytics:** This approach involves analysing textual data like essays, discussion forum posts, and student feedback to extract insights into student learning styles, progress, and challenges [39,40]. Text analytics can identify emerging topics of difficulty within a course, gauge student sentiment towards specific learning materials, or analyse student responses to better understand their thought processes [41].
- **Social Network Analysis:** This technique examines the relationships and interactions between students in a learning environment [42,43]. In analysing collaboration patterns and communication networks, educators can identify isolated students who might benefit from additional connections or foster peer-to-peer learning opportunities [44].
- **Machine Learning Algorithms:** These algorithms can learn from large datasets and identify complex patterns in student data [45]. Machine learning can personalise learning content based on individual student needs, develop adaptive learning systems that adjust difficulty levels, or automatically grade student work [46].
- **Data Visualisation:** Presenting complex data in visually compelling formats helps educators and students gain deeper insights into student performance and learning

patterns [47,48]. Interactive dashboards and reports can be used to track individual or cohort progress, visualise student engagement metrics, and identify areas for improvement [48,49].

While data analytics offers promising applications for enhancing student academic achievement in postgraduate education, its implementation within HEIs typically faces multifaceted challenges and limitations in ensuring student success [50]. Compounding these challenges and limitations, the field of postgraduate education currently lacks standardised guidelines for incorporating data analytics into the learning process [51]. This absence of best practices creates uncertainty for HEIs regarding data collection strategies, effective analysis techniques, and using data-driven insights for programme improvement [52,53]. The inconsistency in data analytics implementation across programmes and institutions can lead to varying levels of effectiveness and impact on student learning outcomes [52,54]. Without clear best practices and adequate support, institutions may not fully leverage the potential of data analytics to enhance student success [55]. In addition, the existing literature on data analytics in postgraduate education primarily focuses on applying individual techniques, such as descriptive analytics or predictive modelling. A gap exists in research that explores the integrated application of diverse data analytics approaches within a postgraduate setting. This integrated approach could offer valuable insights into student learning behaviour and performance, potentially leading to more targeted interventions and improved academic achievement.

To address the gap regarding the efficacy of data analytics in promoting academic success among student populations, this systematic review investigated the integrated application of various data analytics approaches in the postgraduate education context. Thus, by exploring how these combined approaches can be used to gain a more comprehensive understanding of student learning behaviours and academic performance, the review aimed to consolidate existing research on data analytics to support postgraduate students. This consolidation provides valuable recommendations for educators, policymakers, and researchers on optimising data analytics to enhance academic achievement in postgraduate programs. Through a systematic synthesis of a comprehensive corpus of scholarly literature, this review aimed to identify and critically evaluate best practices in employing data analytics to augment postgraduate student academic achievement. This study distinguishes itself by transcending theoretical discourse to offer practical recommendations and actionable strategies for implementing data analytics within the postgraduate educational context. Also, by categorising tactics and suggesting avenues for future research, this review bridges the chasm between theory and practice, thereby providing an indispensable resource for educators and policymakers endeavouring to enhance academic outcomes through data-driven approaches. In this context, the following research objectives guided the review:

1. To explore different approaches used for data analytics in postgraduate education.
2. To examine potential challenges and limitations associated with the application of data analytics in postgraduate education.
3. To establish best practices for practices to cultivate a supportive learning environment for postgraduate students in the context of data-driven educational interventions.

2. Materials and Methods

To ensure transparency and methodological rigour, the researchers adhered to the guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement [56] when conducting this systematic review and meta-analysis.

2.1. Literature Search, Screening, and Application of Eligibility Criteria

A systematic search was performed across five databases, which included ACM Digital Library, ERIC, PsycINFO, Scopus, and Web of Science. The initial search results encompassed peer-reviewed journal articles, conference papers, and chapters. The search

strategy employed a combination of keywords and subject headings related to postgraduate education and data analytics. The following syntax for each database was utilised: ACM Digital Library: ((“postgraduate education” OR “masters” OR “doctoral” OR “graduate education”) AND (“data analytics” OR “learning analytics” OR “educational data mining” OR “predictive analytics” OR “descriptive analytics” OR “prescriptive analytics” OR “text analytics” OR “social network analysis” OR “machine learning algorithms” OR “data visualization”) AND (“student” OR “learner” OR “academic achievement” OR “academic performance”)). ERIC: (TI = (“postgraduate education” OR “masters” OR “doctoral” OR “graduate education”) OR AB = (“data analytics” OR “learning analytics” OR “educational data mining” OR “predictive analytics” OR “descriptive analytics” OR “prescriptive analytics” OR “text analytics” OR “social network analysis” OR “machine learning algorithms” OR “data visualization”)) AND (KW = (“student” OR “learner” OR “academic achievement” OR “academic performance”)). PsycINFO: ((TX = “postgraduate education” OR TX = “masters” OR TX = “doctoral” OR TX = “graduate education”) AND (TX = “data analytics” OR TX = “learning analytics” OR TX = “educational data mining” OR TX = “predictive analytics” OR TX = “descriptive analytics” OR TX = “prescriptive analytics” OR TX = “text analytics” OR TX = “social network analysis” OR TX = “machine learning algorithms” OR TX = “data visualization”) AND (TX = “student” OR TX = “learner” OR TX = “academic achievement” OR TX = “academic performance”)). Scopus: TITLE-ABS-KEY((“postgraduate education” OR “masters” OR “doctoral” OR “graduate education”) AND (“data analytics” OR “learning analytics” OR “educational data mining” OR “predictive analytics” OR “descriptive analytics” OR “prescriptive analytics” OR “text analytics” OR “social network analysis” OR “machine learning algorithms” OR “data visualization”) AND (“student” OR “learner” OR “academic achievement” OR “academic performance”)). Web of Science: TS = ((“postgraduate education” OR “masters” OR “doctoral” OR “graduate education”) AND (“data analytics” OR “learning analytics” OR “educational data mining” OR “predictive analytics” OR “descriptive analytics” OR “prescriptive analytics” OR “text analytics” OR “social network analysis” OR “machine learning algorithms” OR “data visualization”) AND (“student” OR “learner” OR “academic achievement” OR “academic performance”)). Therefore, Boolean operators (AND, OR) were used to combine these terms effectively, enhancing both search precision and recall. To ensure a focused and relevant literature search, this study adopted the Population, Intervention, Comparison, Outcome (PICO) framework for developing the search strategy and establishing eligibility criteria [57]. The PICO framework proved to be a valuable tool in streamlining the initial stages of the study selection process by directing the identification of relevant articles. Following the PICO structure, the eligibility criteria were established as follows:

- Population (P): Postgraduate students (Master’s or Doctoral programmes). Screening: During title and abstract screening, the researchers prioritised studies that explicitly mention postgraduate students as the target population. Excluded studies that focused on primary education, secondary education, undergraduate education, or broader student populations without specifying postgraduate levels.
- Intervention (I): Use of data analytics in the context of postgraduate education. Screening: The researchers looked for studies investigating interventions involving data analytics for student support or learning improvement. Excluded studies solely focused on the technical aspects of data analytics without an educational application (e.g., development of new algorithms).
- Comparison (C): Comparison group or baseline data. Screening: The researchers prioritised studies that compared the effectiveness of data analytics interventions to traditional support methods (e.g., study skills workshops) or a control group with no intervention. Studies with a baseline measure of academic achievement before the intervention were also considered for further review.
- Outcome (O): Postgraduate student academic achievement. Screening: The researchers selected studies that measured academic achievement outcomes relevant to postgrad-

uate students. Examples included grades, graduation rates, time to completion, and thesis/dissertation quality. Building upon the core PICO framework, the following additional eligibility criteria were established:

- Studies published in English to ensure accessibility and consistency in analysis.
- Studies published from January 2016 to July 2024 were considered to ensure the inclusion of the most recent advancements in data analytics within postgraduate education. This timeframe captures the rapid evolution of data analytics technologies and their increasing application in higher education settings. Studies that reported detailed information regarding data analytics interventions implemented. This includes specifics like the structure of the intervention (e.g., frequency of data analysis, feedback loops), the types of data analysed (e.g., student performance data, engagement metrics), and the activities employed to use the data for student support (e.g., personalised learning recommendations, targeted interventions).
- Research articles that demonstrate a strong understanding of key concepts in data analytics and their application in postgraduate education. Additionally, the studies should employ rigorous research methodologies, clearly define the sampling frame of postgraduate students involved, and present valuable findings that contribute to our understanding of how data analytics can enhance student success in this specific educational context.

The PRISMA flow diagram (Figure 2) outlines the search strategy and selection process for included studies.

A total of 2910 studies were identified through the initial literature search. To efficiently manage the vast amount of data and ensure a high-quality selection process, the reviewers (the researchers) implemented a two-stage selection process using both Rayyan (free web-based platform) [58] and ASReview (open-source software) [59] platforms designed to streamline systematic reviews.

- Stage 1: De-duplication and Machine Learning Prioritisation
 - ✓ The reviewers initiated the systematic review process by leveraging Rayyan's advanced deduplication capabilities [58]. This web-based platform employs sophisticated algorithms and machine-learning techniques to identify and eliminate redundant studies [58]. Comparing various attributes such as title, authors, abstract, and publication details, Rayyan accurately flagged 1050 duplicate and non-English studies. This streamlined the review process, significantly reducing manual effort and allowing the reviewers to focus on a more manageable pool of 1860 articles for further evaluation.
 - ✓ Next, the reviewers employed ASReview, a sophisticated machine learning-powered tool designed to streamline the systematic review process [59]. This innovative software automates critical steps, including article screening and data extraction [59]. To initiate the screening process, ASReview analysed the remaining 1860 titles and abstracts based on predefined inclusion and exclusion criteria established within the platform. This analysis prioritised articles for full-text review, focusing on the most promising candidates. Leveraging machine learning, ASReview accurately predicted the relevance of subsequent articles, enabling rapid and reliable screening [59]. This automation significantly accelerated the review process, allowing the reviewers to allocate more time to higher-level analysis and interpretation of the included studies.
- Stage 2: In-depth Review and Selection within Rayyan
 - ✓ Following ASReview's [59] prioritisation, the reviewers proceeded with a meticulous review process using Rayyan [58]. Both reviewers independently assessed the full text of the prioritised articles within the Rayyan interface.
 - ✓ Through rigorously applying the pre-defined criteria established in Rayyan, the reviewers excluded 1815 studies that did not fully meet the eligibility requirements or align with the research objectives. This in-depth evaluation identified forty-five key studies directly addressing data analytics in postgraduate education.

These forty-five studies were retained for a comprehensive quality assessment, focusing on the clarity and comprehensiveness of concepts, methodologies employed, sampling frames, and reported findings.

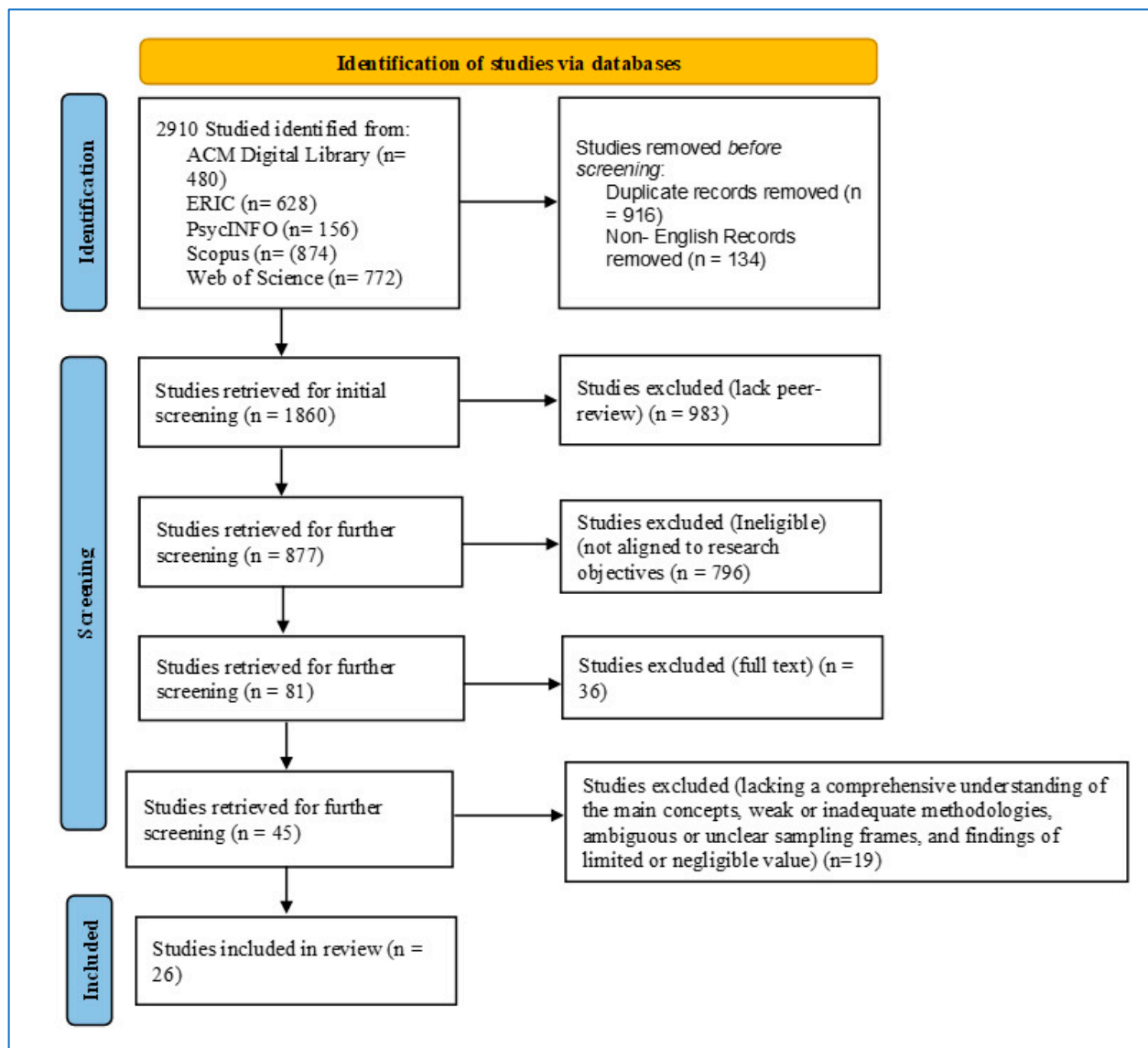


Figure 2. Mapping the Literature Search and Study Selection Process (Adapted from PRISMA Statement [56]).

2.2. Quality Assessment

A comprehensive quality assessment was conducted using the Critical Appraisal Skills Programme (CASP) checklist [60]. The CASP checklist provided a structured framework with specific criteria for assessing each article. This ensured a consistent and standardised approach across all studies, minimising reviewer bias and promoting transparency in the selection process. The checklist directed attention towards four crucial aspects that determine the overall quality and reliability of research articles:

- **Clarity and Comprehensiveness of Concepts:** The CASP checklist prompted the reviewers to evaluate the authors' understanding of core concepts in data analytics and their application to postgraduate education [60]. The reviewers assessed the introduction and literature review sections for evidence of a clear grasp of relevant theories, prior research, and the specific challenges and opportunities in this context.

- **Rigorous Methodologies:** The checklist guided the scrutiny of the methodology sections. The reviewers examined aspects like the appropriateness of the research design for the research questions, data collection methods (surveys, interviews, etc.), data analysis techniques, and the researchers' attention to potential bias and its mitigation. Studies employing well-justified and rigorous methodologies were prioritised [60].
- **Clear Sampling Frames:** The CASP checklist [60] emphasised the importance of clear sampling frames. The reviewers evaluated whether the authors clearly defined the target postgraduate student population, the sampling method used to select participants, and the justification for the sample size and its representativeness. Studies with well-defined and representative sampling frames were considered more reliable.
- **Valuable Findings:** The checklist directed focus to the results and discussion sections. The reviewers assessed the clarity of the presented findings, their alignment with the research questions and methodology, and the depth of discussion regarding their meaning and implications. Articles presenting valuable contributions to the knowledge base on data analytics in postgraduate education, along with practical recommendations, were deemed particularly noteworthy [60].

Using the CASP checklist [60] as a guiding framework ensured a systematic and thorough evaluation of each article's strengths and weaknesses across these four key quality indicators. This approach led to the selection of a collection of high-quality research articles that formed the foundation for the comprehensive review. Following this assessment, a further nineteen studies were excluded for not meeting the predetermined quality criteria. The final review stage comprised a comprehensive analysis and synthesis of the research findings from the remaining twenty-six high-quality studies.

Following the initial stage of independent article screening using Rayyan and ASReview software, reviewers convened to discuss and resolve any discrepancies that emerged during the selection process. This approach ensured a balanced and objective evaluation of each study's relevance to the review. To confirm the consistency of the screening process and minimise bias, the reviewers established measures of inter-rater reliability (IRR). The percentage of agreement between the reviewers served as a primary indicator, with a high concordance rate of 90% achieved. Furthermore, the reviewers calculated the kappa statistic (κ), a more robust measure that accounts for chance agreement. The resulting kappa coefficient of 0.85 suggested a strong level of agreement between the reviewers, further solidifying the reliability of the screening process. These measures demonstrate the rigour employed to ensure a comprehensive and unbiased selection of studies for this review.

2.3. Data Extraction and Coding

Data extraction was conducted following the guidelines outlined in the PRISMA checklist, ensuring adherence to best practices in systematic review methodology. For the standardised data extraction process, the researchers used an open-access online tool called CADIMA. This tool facilitated the systematic extraction of pertinent information from the selected studies, promoting consistency and efficiency in the data extraction process [61]. The data extraction form included the following elements:

- Study characteristics (e.g., author, year of publication, geographical location).
- Research design and methodology.
- Data analytics techniques and tools used.
- Key findings related to the benefits and challenges of data analytics in postgraduate education.
- Limitations or areas for further research identified by the authors.

Data coding in this systematic review was performed manually. It involved extracting and documenting essential information such as author(s), research design, publisher, and findings. The extracted data were then organised and presented in Appendix A, which provides a comprehensive overview of the data extraction and coding process employed in this review. Table 1 provides a brief overview of the studies.

Table 1. Overview of Reviewed Studies.

Author(s)	Focus
[36]	Introducing prescriptive and predictive analytics to MBA Students with Microsoft Excel
[42]	International doctoral students' academic socialisation in China: A social network analysis
[50]	Learning analytics to develop future competences in higher education: a case study
[62]	Exploring the use of educational data mining and learning analytics to improve instructional practices and student performance
[63]	Predicting master's students' academic performance: An empirical study in Germany
[64]	Data in academia: A proposed framework for improving student's performance
[65]	Predict or describe? How learning analytics dashboard design influences motivation and statistics anxiety in an online statistics course
[66]	Predicting student performance in a Master of Data Science program using admissions data
[67]	Adopting learning analytics to inform postgraduate curriculum design: Recommendations and research agenda
[68]	Human–AI collaboration patterns in AI-assisted academic writing
[69]	Predictive analysis of motivation and learning strategies on academic achievement of postgraduate students
[70]	Analysis of students' academic performance by using machine learning tools
[71]	Peer-mentoring and doctoral student retention: a longitudinal investigation
[72]	Analysis of text mining from full-text articles and abstracts by postgraduate students in selected Nigeria universities
[73]	Student perceptions of privacy principles for learning analytics
[74]	Predictive learning analytics and the creation of emotionally adaptive learning environments in higher education institutions: A study of students' affect responses
[75]	Influence of social media on academic achievement of students of the Central University of Punjab
[76]	Teaching tip: Using text analytics AI insights in Microsoft Power BI Desktop to score sentiments, extract key phrases, and discover unstructured data patterns
[77]	Visualisation and data analysis of multi-factors for the scientific research training of graduate students
[78]	Supporting students' academic performance using explainable machine learning with automated prescriptive analytics
[79]	Dispositions towards learning: the importance of epistemic attributes for postgraduate learners
[80]	Single-case learning analytics: 784 Feasibility of a human-centered analytics approach to support doctoral education
[81]	Impact of learning analytics guidance on student self-regulated learning skills, performance, and satisfaction: A mixed methods study
[82]	Assessing the impact of digital education and the role of the big data analytics course to enhance the skills and employability of engineering students
[83]	Learning analytics in collaborative learning supported by Slack: From the perspective of engagement
[84]	Learning analytics framework based on human-centered artificial intelligence for identifying the optimal learning strategy to intervene in learning behavior

3. Results

This section delves into a critical analysis of the literature review findings in the context of the established research objectives. The initial phase of this review involved categorising the studies according to their learning environment. This was a crucial step in understanding the broader context of optimising data analytics to enhance postgraduate student academic achievement. Thus, by identifying the dominant learning environments employed in the studies, it was possible to analyse the specific challenges and opportunities associated with each approach and to draw meaningful conclusions about the effectiveness of data analytics in different educational settings. Table 2 presents the classification results.

Table 2. Application of data analytics online and blended environments.

Learning Environment	Studies	Benefits
Online	[36,42,65,66,68,70,72,75–78,80,82–84]	Descriptive Analytics: Identify struggling students, and track progress remotely.
		Text Analytics: Gain insights into student understanding to inform targeted interventions.
		Predictive Analytics: Proactively identify at-risk students and provide early intervention.
		Prescriptive Analytics: Recommend specific interventions to address skill gaps.
		Machine Learning Algorithms: Create personalised learning paths to address individual needs.
		Data Visualisation: Track progress and identify trends to inform instruction.
Blended	[50,62–64,67,69,71,73,74,79,81]	Descriptive Analytics: Identify struggling students in both online and face-to-face settings, track progress, and tailor instruction accordingly.
		Text Analytics: Gain insights into student understanding to inform targeted interventions in both online and face-to-face settings.
		Predictive Analytics: Proactively identify at-risk students in both online and face-to-face settings and provide early intervention.
		Prescriptive Analytics: Recommend specific interventions to address skill gaps in both online and face-to-face settings.
		Machine Learning Algorithms: Create personalized learning paths to address individual needs in both online and face-to-face settings.
		Data Visualisation: Track progress and identify trends to inform instruction in both online and face-to-face settings.

Table 2 demonstrates the potential of data analytics to significantly enhance learning outcomes in both online and blended learning environments by providing educators with valuable insights and tools to support student success. While the reviewed studies did not specifically focus on the face-to-face learning environment, the analysis of online and blended learning environments offers valuable insights. To further delve into the application of data analytics in postgraduate education, the studies were systematically categorised based on educational level and data analytics approach. This classification enabled the identification of patterns and trends, leading to the comprehensive overview presented in Table 3.

Table 3 provides a comprehensive overview of data analytics approaches employed by graduate students, including those pursuing master’s and doctoral degrees, highlighting the practical applications of blended learning in enhancing teaching and learning experiences. While the category ‘Graduate Students’ encompasses both levels, it is important to note that the studies within this category did not differentiate between master’s and doctoral students. This table also explores how data analytics can improve education in four key areas: foundational skills, motivation and engagement, effective teaching and learning strategies, and assessment and evaluation. For each area, the table lists relevant scholarly sources, describes specific data analytics approaches, and details the potential benefits for educators and students. Building upon the analysis of data analytics approaches, Table 4 delves into the key challenges and best practices associated with their implementation in postgraduate education.

Table 4 examines best practices for incorporating data analytics into postgraduate education. It identifies several challenges, including data privacy concerns, potential biases, and a lack of data literacy among faculty and students. The table then offers solutions, emphasising faculty training, strategic technology use, and fostering collaborative learning environments that prioritise transparency and ethical considerations.

Table 3. Application of data analytics approaches in postgraduate education.

Education Level	Data Analytics Approach	Studies	Key Aspects and Processes	Application
Masters	Predictive Analytics	[63–66,74]	Foundational Skills; Motivation and Engagement; Effective Teaching and Learning Strategies; Assessment and Evaluation	Proactively identify at-risk students, provide targeted interventions, optimise teaching approaches, and enhance student success.
	Descriptive Analytics	[65,67,79]	Foundational Skills; Effective Teaching and Learning Strategies; Assessment and Evaluation	Identify areas of struggle, inform targeted interventions, and focus instruction on areas of need.
	Machine Learning Algorithms	[74]	Foundational Skills; Motivation and Engagement; Effective Teaching and Learning Strategies	Create adaptive learning experiences, increase student engagement, and tailor instruction to individual needs.
	Prescriptive Analytics	[36,50]	Foundational Skills; Effective Teaching and Learning Strategies; Assessment and Evaluation	Provide targeted interventions, optimise teaching approaches, and enhance student success.
	Text Analytics	[72,76]	Foundational Skills; Motivation and Engagement	Identify areas of lack of understanding, inform targeted interventions, and gauge motivation.
	Social Network Analysis	[83]	Motivation and Engagement	Foster a collaborative learning environment, increase student engagement
Doctorate	Social Network Analysis	[42]	Motivation and Engagement	Foster a collaborative learning environment, increase student engagement
	Data Visualisation	[77,80,82]	Motivation and Engagement	Motivate students by highlighting progress and achievements
	Machine Learning Algorithms	[68]	Foundational Skills; Motivation and Engagement; Effective Teaching and Learning Strategies	Create adaptive learning experiences, and increase student engagement.
	Descriptive Analytics	[71]	Foundational Skills; Motivation and Engagement; Effective Teaching and Learning Strategies	Identify areas of struggle, and inform targeted interventions.
Graduate Students	Prescriptive Analytics	[78]	Foundational Skills; Motivation and Engagement, Effective Teaching and Learning Strategies; Assessment and Evaluation	Provide targeted interventions, optimise teaching approaches, and enhance student success.
	Descriptive Analytics	[62,73,75]	Foundational Skills, Motivation and Engagement; Assessment and Evaluation	Identify areas of struggle, inform targeted interventions, and focus instruction on areas of need.
	Predictive Analytics	[69,81]	Foundational Skills; Effective Teaching and Learning Strategies; Assessment and Evaluation	Proactively identify at-risk students, provide targeted interventions, and optimise teaching approaches.
	Machine Learning Algorithms	[70,84]	Foundational Skills; Motivation and Engagement; Effective Teaching and Learning Strategies	Create adaptive learning experiences, increase student engagement, and tailor instruction to individual needs.

Table 4. Challenges and Best Practices in Postgraduate Education.

Themes	Studies	Findings
Challenges and Limitations of Data Analytics in Postgraduate Education	[67,82]	Heterogeneity of assessment: Assessing student learning in postgraduate programmes is complex due to the variation in disciplines and expertise. Standardised tests, often designed for broader subject areas, may struggle to capture the nuanced knowledge and skills acquired through a programme’s specific curriculum.
	[63,67,71,72,75]	Knowledge translation gap: Integrating data into postgraduate education presents a multifaceted challenge. While organisational and cultural barriers like resistance to change and user training exist, a potentially more critical hurdle lies in fostering a data-driven culture. Postgraduate programmes often cultivate faculty with deep disciplinary knowledge, yet communication skills, particularly those focused on translating knowledge into clear, data-driven formats, may be less emphasised.
	[72,73,75]	Data privacy challenges: Data privacy throws a wrench into data analytics for postgraduate education. Sensitive student data like grades and research findings are subject to strict regulations. Anonymising data can limit its usefulness while obtaining informed consent and ensuring ethical use add complexity. Even anonymised data can perpetuate biases, requiring careful data selection and model monitoring.
	[70,78,79,84]	Algorithmic bias: The potential for bias lurks within both data and algorithms used for data analytics in postgraduate education. Data itself might reflect societal biases, like historical correlations between gender and grades (due to external factors). Algorithms can also introduce bias if not carefully designed. For instance, an algorithm predicting student success based on past performance might overlook factors like socioeconomic background.
	[42,67,72,73,78]	Data literacy gap: Data literacy is a critical but neglected skill in postgraduate education. While data analytics offers exciting possibilities, it requires individuals to understand, analyse, and communicate data effectively. However, a theory–practice gap exists. Programmes might teach data analytics concepts but lack resources for practical skill development. Similarly, educators might not have these skills themselves.
	[72,73,75]	Resource constraints in data analytics: Data analytics can be expensive. Setting up and maintaining the needed infrastructure like high-powered computers and storage is costly. Many educational institutions lack the resources for this, which hurts data quality and relevance for analysis. Limited computing power might not manage large datasets, leading to incomplete insights. Insufficient storage makes it hard to keep historical data, crucial for tracking trends.
	[67,71,78,79]	Data interpretation complexity: In postgraduate education, the successful application of data analytics hinges on the ability to interpret complex and multifaceted datasets accurately. These datasets typically encompass a range of student performance data, course evaluations, demographic information, and potentially qualitative feedback. However, ensuring accurate interpretation presents a multifaceted challenge. Inaccurate or incomplete data, such as missing values or inconsistencies, can lead to misleading or unreliable findings. Educators or analysts lacking a deep understanding of the educational context and the specific nuances of the data might misinterpret patterns or trends.
Best Practices for Ensuring a Supportive Learning Environment for Data Analytics Application in Postgraduate Education	[63,67,78]	Data literacy development: Prior to implementing curriculum adaptations focused on data-driven approaches, equipping educators and administrators with foundational skills in data collection and management is essential.
	[36,62,65,67,68,70,78,80,84]	Bridging the theory–practice gap with technology integration: Postgraduate data analytics education can bridge the theory–practice gap through strategic technology integration. Cloud platforms like Amazon Web Services (AWS) and Google Cloud Platform (GCP) offer access to vast datasets, exceeding the limitations of traditional classrooms. This allows students to transition from theoretical frameworks to real-world data exploration. Furthermore, open-source tools like R, Python, and Structured Query Language (SQL) equip students with industry-standard skills for data manipulation, analysis, and visualisation. Active engagement with these tools fosters a deeper understanding of data analytics concepts and prepares graduates for data-driven careers by ensuring they are proficient in the tools they will encounter in the workforce.

Table 4. Cont.

Themes	Studies	Findings
Best Practices for Ensuring a Supportive Learning Environment for Data Analytics Application in Postgraduate Education	[64,70,72,74,78,79,84]	Collaborative learning and industry partnerships: Fostering collaboration and peer learning are crucial aspects of postgraduate data analytics education. Two key strategies can achieve this. Collaborative projects capitalise on the diverse skills needed for data analysis, bringing students together to leverage each other's strengths in data collection, cleaning, analysis, and interpretation. This promotes teamwork, communication, and a deeper understanding of the entire data analytics process. Industry partnerships provide access to real-world datasets, more complex than classroom examples. By tackling these practical problems, students bridge the theory–practice gap, learn from industry professionals' perspectives, and gain valuable experience for their data-driven careers.
	[62,72,73,75,84]	Prioritising transparency and ethical discussions: Open communication about data practices builds trust and addresses privacy concerns. Discussions about bias and responsible data use help mitigate bias and ensure fairness. Informed consent empowers students regarding their data while fostering data literacy equips them to critically evaluate data-driven insights.
	[62,68,80,84]	Lifelong learning in data analytics: Cultivating a continuous learning mindset is essential for postgraduate students navigating the challenges of data analytics. This rapidly evolving field demands critical thinking skills to stay abreast of the latest advancements in techniques, tools, and technologies. Hence, through actively seeking out new knowledge and engaging with emerging trends, students can not only improve their technical skills but also develop the critical thinking necessary to effectively apply data analytics in various contexts. This ongoing learning fosters adaptability and empowers students to overcome challenges and navigate the complexities of the data analytics landscape throughout their careers.
	[36,50,62,65–68,71,72,74,81,82,84]	Faculty development and data-driven assessment: The data literacy gap between faculty and student needs requires a multifaceted solution. Faculty development programmes can bridge this gap by offering training in areas like statistical analysis and programming languages (Python, R). This empowers faculty to not only integrate data analytics into teaching but also utilise diverse assessment methods beyond exams. These might include data-driven assignments, portfolio analysis, or peer evaluations, allowing for a more holistic evaluation of student learning outcomes informed by data. Hence, by developing faculty expertise and using diverse assessments, postgraduate education fosters a data-literate culture, creating a more effective and data-driven learning environment for all.

4. Discussion

The analysis of Table 2 indicates a prevailing trend towards online and blended learning environments in the reviewed studies, suggesting a growing emphasis on flexibility and technology-enhanced pedagogy in graduate education [8]. The increasing adoption of blended learning approaches highlights the potential benefits of integrating both online and in-person components to address diverse student needs and enhance learning outcomes [9,10]. This concurs with the assertions of Hill and Smith [10] and Soncin et al. [11], who posit that blended learning approaches, integrating face-to-face instruction with online learning platforms, can cater to diverse learning preferences and provide access to specialised resources. Building on this understanding, Table 3 provides a comprehensive overview of the data analytics approaches employed in graduate studies, specifically at the master's and doctoral levels, thereby highlighting the practical applications of blended learning in enhancing teaching and learning experiences. Although the category “Graduate Students” encompasses both master's and doctoral students, it is essential to note that studies within this category did not differentiate between these specific levels. A notable finding is the predominance of master's level research, which accounts for the majority of studies ($n = 11$). The results also suggest a widespread adoption of data analytics techniques at this level [63,66]. Master's students employ a diverse range of approaches, including predictive analytics, descriptive analytics, machine learning algorithms, prescriptive analytics, text analytics, and social network analysis.

This diversity indicates the broad applicability of data analytics in addressing various research questions and challenges within master's programmes [63]. These findings align with Lohle's [85] study, which highlighted the potential for master's students to alleviate anxiety related to technical content by effectively utilising analytics. In contrast, doctoral-level research tends to focus on specific data analytics approaches. The prevalence of social network analysis and data visualisation, machine learning algorithms and descriptive analytics techniques at the doctoral level indicates a deeper exploration of complex research problems [42,71,80]. In accord, Jiang et al. [43] noted that these approaches allow researchers to investigate intricate relationships between variables, identify patterns and trends, and communicate findings effectively. In alignment with House's [86] findings, the scarcity of studies categorised as postgraduate diploma programmes might be attributed to factors such as limited research, insufficient integration of data analytics in curricula, or constraints in data availability.

Table 3 unveiled an array of data analytics approaches, which empower educators to strategically leverage student data for diverse purposes [78]. The review found that this includes identifying specific student needs through techniques like descriptive analytics and text analytics [67,78]. The review also found that text analytics relating to open-ended responses on assignments, for instance, can pinpoint areas where students might lack basic understanding [79]. Armed with this knowledge, the review found that educators can design personalised learning experiences that effectively strengthen those skills [36]. In accordance with these findings, Fernandes et al. [4], Saadati et al. [5], Saeed and Al Qunayeer [9], Mishra et al. [37] and Jagwani and Aloysius [45] underscore the need to employ methodologies capable of uncovering latent student misconceptions and knowledge deficits through the analysis of written assignments. Such insights are instrumental in formulating targeted pedagogical interventions to enhance student learning outcomes. The review highlighted the utility of social network analysis, text analytics, machine learning algorithms, and data visualisation for fostering a more effective and engaging learning environment [78]. Thus, the review established that social network analysis can identify isolated students who might benefit from peer learning opportunities [70]. While the literature review revealed that text analytics of student sentiment expressed within course forums and surveys can be instrumental in assessing student motivation and pinpointing areas of dissatisfaction, [79]. As such, by addressing these concerns, the review revealed that educators can cultivate a more collaborative learning environment and develop strategies to increase intrinsic motivation for learning [75]. What also emerged from the literature review is that machine learning algorithms personalise learning content and suggest relevant learning pathways based on student progress and interests [42]. The review also established data visualisation as another key technique that can create engaging dashboards to visualise student progress and achievements, fostering a sense of accomplishment [78]. Therefore, the review underscored that these approaches could increase student engagement by making learning experiences more relevant and interesting while motivating students by highlighting their progress in a visually compelling way [71,79]. In synch with these findings, Abdul-Jabbar and Farhan [24], Sivarajah et al. [26], Pinto et al. [46] and Kleimola and Leppisaari [50] elucidated that the integration of social network analysis, text mining, machine learning, and data visualisation techniques within educational contexts has the potential to revolutionise teaching and learning by providing valuable insights into student behaviour, learning patterns, and social interactions.

The review, through Table 3, emphasised the value of data analytics techniques such as descriptive analytics, text analytics, predictive analytics, prescriptive analytics, machine learning algorithms, and data visualisation [67,78]. The review noted that these approaches enable educators to identify the most effective teaching methods for specific learning objectives and diverse student populations [67]. Also, the literature review revealed that descriptive analytics of student performance data across different teaching methods, for example, can inform the development of more effective teaching materials [78]. Concerning predictive analytics, the review noted that these can go a step further by identifying teaching methods that are most likely to be successful for different student groups based

on historical data [84]. Also, prescriptive analytics can then recommend specific teaching strategies or resources based on these models and student learning needs [64]. The review underlined that this empowers educators to optimise teaching approaches to maximise student learning outcomes for diverse learners and provide data-driven guidance on selecting the most effective methods [53,81]. The literature review established that machine learning algorithms can be harnessed through adaptive learning systems that adjust difficulty levels and provide real-time personalised feedback based on student performance [36]. This tailors instruction to individual student needs and paces, ensuring optimal learning experiences [25]. Finally, through Table 3, the review underscores the use of data analytics in assessment and evaluation. Descriptive analytics and text analytics of student responses to assessments can be used to identify learning gaps and areas for improvement [65]. This allows educators to focus instruction on areas where students struggle and refine assessments to better gauge student understanding [69]. The review also noted that predictive analytics can further identify students at risk of performing poorly on upcoming assessments, enabling educators to proactively implement interventions and bolster student success [72]. Consequently, the multifaceted applications of data analytics offer a transformative potential for enhancing teaching and learning [32,34,36,37], by optimising instructional strategies, personalising learning experiences, and ultimately improving student outcomes [39,42,45,47].

A comprehensive analysis of Table 4 within this literature review unveiled both the promise and pitfalls inherent in integrating data analytics into postgraduate education. A prominent challenge lies in the limitations of standardised tests [67]. The review established that these instruments, often designed for broader subject areas, struggle to capture the nuanced knowledge and competencies acquired through the diverse range of postgraduate programmes, potentially overlooking valuable skills and expertise [67]. The review noted that this necessitates a more comprehensive approach to student evaluation in postgraduate settings, potentially involving faculty input, portfolio assessments, or tailored evaluations specific to the programme's focus [73]. Furthermore, Table 4 highlights the complexities of data integration within postgraduate education [67]. The literature review noted that obstacles include overcoming organisational and cultural resistance to change [67], fostering a data-driven culture that emphasises clear communication of findings to diverse stakeholders [63], and addressing potential shortcomings in faculty communication skills. Postgraduate programmes often involve faculty with strong disciplinary expertise, but less emphasis on translating that knowledge into clear, data-driven communication [67]. An additional challenge identified in Table 4 pertains to data privacy regulations [75]. The review noted that stringent regulations governing sensitive student data, such as grades and research findings, pose a significant hurdle. Thus, balancing the need for anonymisation to protect student privacy, informed consent to ensure students understand how their data are being used, ethical data use practices, and student data ownership rights creates complexities in data management practices [72,75]. Therefore, there are multifaceted challenges and opportunities associated with integrating data analytics into postgraduate education [50]. The limitations of standardised assessments in capturing the intricacies of postgraduate learning, coupled with the complexities of data integration, including organisational resistance, data privacy concerns, and the need for enhanced data literacy [50–52,54], highlight the need for a nuanced and multifaceted approach to harnessing the potential of data analytics in this context [55].

Building on the challenges identified, this review explored key strategies from Table 4 for fostering a supportive environment for data analytics in postgraduate education. Equipping educators and administrators with core data collection and management skills was found to be crucial [68], as highlighted in Table 4. This empowers them to confidently implement data-driven curriculum adaptations [67]. Table 4 emphasises the value of cloud computing platforms and open-source tools like R, Python, and SQL. These resources bridge the gap between theoretical knowledge and practical application [65,66]. Hence, by providing access to vast datasets and industry-standard tools for data manipulation,

analysis, and visualisation, they empower students to engage with real-world data challenges from the outset [78]. Another best practice identified in this review (Table 4) involves collaborative projects and industry partnerships [42,83]. The review underlined that these approaches promote teamwork, communication, and a deeper understanding of the entire data analytics process. Students gain invaluable experience by working with diverse skill sets and tackling real-world data challenges alongside industry professionals [64]. Open communication about data practices fosters trust and addresses privacy concerns [75]. Additionally, fostering discussions about bias and responsible data use helps mitigate potential biases within the data or algorithms [72]. Informed consent empowers students regarding their data while cultivating data literacy, equipping them to critically evaluate data-driven insights and ensure responsible use of their data [79]. Consistent with the findings of this review, Hassna [55] emphasised that institutions can establish a robust foundation for data analytics integration in postgraduate education by effectively addressing the inherent challenges and limitations within their respective contexts.

The data analytics landscape is constantly evolving. As such, the review revealed that cultivating a continuous learning mindset is essential for postgraduate students [67]. Critical thinking skills were also found to be necessary in staying updated on the latest advancements in techniques, tools, and technologies [5]. Hence, by actively seeking new knowledge and engaging with emerging trends, students refine their technical skills and adaptability, allowing them to navigate the complexities of data analytics throughout their careers [84]. Table 4 also emphasises the significance of faculty development programmes [13]. Equipping faculty with expertise in statistical analysis and programming languages (Python, R) empowers them to integrate data analytics effectively into their teaching. This fosters a data-literate culture and creates a more effective, data-driven learning environment for all postgraduate students [62,68]. Through implementing these best practices established in this review (Table 4), postgraduate institutions can establish a supportive environment for data analytics. This paves the way for enhanced student learning, improved teaching strategies, and the creation of a more effective data-driven learning ecosystem [62,68]. In line with the results of this review, Bozkurt and Sharma [28] and Nguyen et al. [29] argue that the dynamic nature of the data analytics landscape necessitates a lifelong learning approach for both students and faculty.

Building upon the preceding comprehensive review of data analytics approaches, their inherent limitations, and optimal strategies for cultivating a conducive learning environment, Figure 3 provides a consolidated summary of the study's outcomes regarding data analytics integration within postgraduate education. This overview underscores a data-informed enhancement process for postgraduate programmes. Continual assessment and refinement are pivotal in optimising academic outcomes. Through the application of diverse data analytics techniques to student data, educators can pinpoint areas of academic strength, weakness, and opportunity. This iterative approach facilitates the creation and implementation of targeted interventions, thereby cultivating a supportive learning ecosystem that elevates student success.

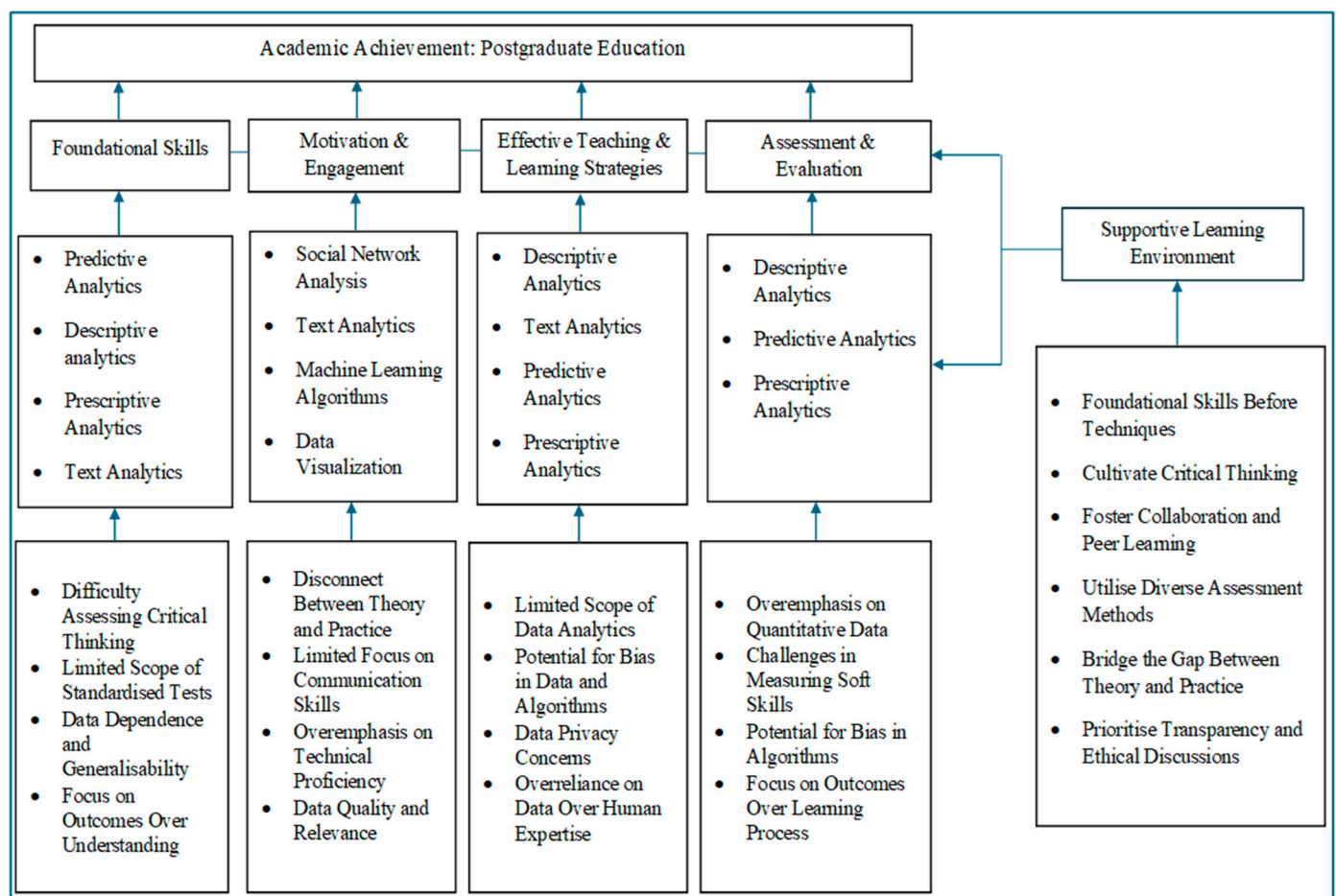


Figure 3. Optimising data analytics to enhance postgraduate student academic achievement.

5. Conclusions

In conclusion, this study underscores the transformative potential of data analytics in reshaping postgraduate education. Thus, by employing a range of techniques, including descriptive, predictive, prescriptive analytics, text analytics, social network analysis, machine learning, and data visualisation, educators can gain valuable insights into student learning. These insights can be used to enhance student support, optimise teaching strategies, and improve assessment practices. However, challenges such as data privacy, faculty development, and standardised assessment limitations must be addressed. As such, through implementing effective strategies, including data literacy training, technology integration, and collaborative partnerships, institutions can create a supportive environment for data-driven decision-making, ultimately leading to improved student outcomes. The results and recommendations from this review can inform educational institutions in making informed decisions about the implementation and utilisation of data analytics tools and strategies. Thus, through leveraging the power of data analytics, postgraduate education can be enhanced, leading to improved learning outcomes and better preparation of students for their professional careers.

6. Limitations of the Present Study

This systematic review prioritised evidence quality by centring on peer-reviewed studies. However, this approach may have inadvertently excluded valuable unpublished research (grey literature) on data analytics and postgraduate student academic achievement. This limitation could have introduced publication bias, favouring statistically significant positive findings. Additionally, the English-language restriction may have limited the inclusion of relevant non-English studies. To mitigate these limitations, the review used a

multi-database search strategy to maximise the identification of relevant peer-reviewed studies. Explicit inclusion and exclusion criteria were applied to enhance the transparency of the selection process. While the specific selection criteria may limit the generalisability of findings to all postgraduate settings, a detailed description of these criteria enables researchers to assess the applicability of the findings to their specific contexts.

7. Future Research Directions

While the current review focused on analysing existing research, future studies could build upon this foundation by conducting empirical research that investigates the practical application of data analytics in diverse postgraduate programs. This could involve case studies examining the implementation process, the effectiveness of specific techniques, and challenges encountered in real-world settings. Additionally, quantitative research methods could be employed to explore the relationships between data analytics interventions and various student outcomes, such as academic achievement, engagement, and satisfaction. While this review provided an overview of diverse data analytics approaches, other researchers could delve deeper into the efficacy of specific techniques within the context of postgraduate education. This could involve longitudinal studies tracking the impact of a particular technique (e.g., learning analytics dashboards, social network analysis) on student learning over time. Additionally, comparative studies could explore the relative effectiveness of different techniques for achieving specific learning objectives within different postgraduate programs. In addition, to gain a more holistic understanding, further research could employ qualitative methods to explore the lived experiences of students and educators involved with data analytics initiatives in postgraduate programs. Semi-structured interviews and focus groups could be used to capture the perspectives of stakeholders on the impact of data analytics on teaching, learning, and the overall learning environment. This would provide valuable insights into potential challenges, ethical considerations, and areas for improvement in data analytics implementation.

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Appendix A Data Extraction and Coding

Author(s)	Design	Publisher	Findings
[36]	The research design involved an application-focused MBA course utilising Microsoft Excel and mathematical programming	INFORMS Transactions on Education	The study focused on a new MBA course that combines prescriptive and predictive analytics with business applications. It used a flipped classroom with real-world examples, equipping students to not only use models but also communicate insights and navigate ethical considerations. The findings suggested that this approach effectively prepares “data-savvy” managers for complex projects

Author(s)	Design	Publisher	Findings
[42]	A qualitative case study design incorporating social network analysis	Research Papers in Education	The research highlighted the complexity of the students' social networks, demonstrating that interactions extended beyond just supervisors. This finding underscores the importance of considering the broader social context for understanding academic socialisation.
[50]	Employed a qualitative case-based research design, utilising group interviews and qualitative content analysis	International Journal of Educational Technology in Higher Education	This study highlights the importance of future competencies like self-awareness, learning literacy, and agency for students. Learning analytics was viewed as a promising tool to foster these competencies through self-reflection, goal setting, and promoting social interactions in learning environments.
[62]	A quantitative study using a Likert scale and random sampling of collegiate students	Journal of Data Acquisition and Processing	This research highlights the multifaceted potential of learning analytics in optimising educational outcomes. By analysing student data, educators can proactively support struggling students, refine teaching based on learning effectiveness, and improve assessment and feedback. Additionally, learning analytics informs institutional decisions on resource allocation and policy development. This data-driven approach has the potential to transform education by personalising learning experiences and maximising student success.
[63]	Data mining algorithms on a limited dataset of easily accessible features from students	Smart Learning Environments	This study identified both academic background (GPA) and non-cognitive factors (motivation, time management, self-regulation) as significant predictors of master's student success. Gender and age were not found to be relevant factors. The findings emphasise the need for a holistic approach that considers both academic merit and non-cognitive skills when assessing and supporting master's students.
[64]	A manual experiment was conducted on 2026 postgraduate students	Revue d'Intelligence Artificielle	This study highlights both the potential and limitations of big data in learning analytics. While the vast amount of micro-level data allows for granular analysis, it can miss broader patterns. The authors propose an approach for analysing data from three categories (location, prior subjects, gender) to recommend subjects to students. This initial experiment showed positive results with students' improved performance.
[65]	This study compared the effects of predictive and descriptive dashboards on learners' motivation and statistical anxiety in a data science course using questionnaires and interviews.	Educational Technology Research and Development	This study found that a predictive dashboard significantly reduced learners' interpretation anxiety and influenced intrinsic goal orientation based on initial levels. Both predictive and descriptive dashboards reduced anxiety for learners with high initial levels. Thematic analysis revealed that performance-avoidance goal orientation was associated with higher anxiety levels, regardless of dashboard usage.
[66]	This study employed a retrospective data analysis approach to predict the academic performance of Master of Data Science applicants.	International Educational Data Mining Society	This study utilized data mining to predict academic success in a Master of Data Science program. In analysing admissions data, key factors influencing performance were identified and predictive models were developed. The results demonstrate the models' accuracy in predicting student success, suggesting their potential for improving admissions decisions.

Author(s)	Design	Publisher	Findings
[67]	A four-year longitudinal study capturing sentiments of postgraduate students at a university in Ireland	Information Systems Frontiers	The researchers provided recommendations and outlined a research agenda for incorporating learning analytics into the design of postgraduate curricula. They emphasised the importance of using data-driven insights to improve teaching and learning outcomes in higher education. Their findings highlighted the potential benefits of using learning analytics in curriculum design, such as identifying student needs, improving retention rates, and enhancing overall learning experiences.
[68]	The research conducted a comprehensive analysis of the characteristics and qualities that enable teachers to effectively integrate technology into instruction	Journal of Research on Technology in Education	The findings suggest that teacher technology change is a complex and multifaceted process influenced by several factors, including knowledge, confidence, beliefs, and cultural factors. Understanding these intersections can inform efforts to support and facilitate effective technology integration in educational settings.
[69]	A correlational research design using the Motivated Strategies for Learning Questionnaire on a sample of 333 participants selected through multi-stage sampling	International Journal of Social Sciences and Education Research	The analysis yielded statistically significant positive associations between motivational and learning strategies and postgraduate students' academic achievement. This suggests that both motivational factors and the use of specific learning strategies independently contribute to academic success. Additionally, the findings indicated a synergistic effect, implying that the combined influence of motivation and learning strategies is even more potent in predicting academic achievement.
[70]	The study developed Python tools for data management, defined academic success criteria, and built predictive models using machine learning and neural networks	Advances in Social Science, Education and Humanities Research	The study found that data mining techniques offer significant advantages in constructing information-analytical systems. These systems extend beyond traditional data modelling and visualisation by enabling the prediction of stable trends.
[71]	A longitudinal design was employed in a Canadian university. Data were collected from 35 participants over four time points.	Higher Education Research and Development	Data analysis revealed a positive association between high levels of peer mentoring and sustained autonomous motivation among students, coupled with low intentions to leave the programme.
[72]	A quantitative survey design was employed, administering a questionnaire to a sample of 357 postgraduate students selected using Raosoft sample size calculation	International Journal of Higher Education	The study uncovered several interesting aspects of how postgraduate students use text mining. For their dissertations and research seminars, students primarily mined full-text articles. Text mining was also used for personal academic development. Abstracts were another source of mined text, although for dissertations and seminars. In terms of the techniques used, information extraction, retrieval, and summarisation were the most common.
[73]	An exploratory study investigated 330 students' perceptions of learning analytics	Journal of Learning Analytics	The study highlighted the importance of considering students' perceptions and concerns about privacy principles in the design and implementation of learning analytics systems.
[74]	A qualitative study involved showing 42 Northern Irish students	International Journal of Information and Learning Technology	The study revealed a heterogeneity in student responses to the predictive learning analytics (PLA) outputs. This spectrum of affective states included curiosity and motivation, comfort and scepticism, confusion and fear, alongside disinterest and doubt regarding the predictions' accuracy. Notably, the authors highlight that not all PLA-induced emotional responses necessarily translate into desirable or productive learning behaviours.

Author(s)	Design	Publisher	Findings
[75]	Descriptive survey research design was employed to gather data from a sample of 100 students selected using stratified random sampling	Research and Reflections on Education	The study's findings indicated that a sizeable portion of postgraduate students at the Central University of Punjab exhibited signs of social media addiction. However, the research revealed a positive association between social media use and academic achievement among this population. This association is linked to the availability of educational resources and study materials online, which students utilise to enhance their academic performance.
[76]	A quasi-experimental design was used to assess the effectiveness of using Microsoft Power BI Desktop	Journal of Information Systems Education	The study involved students new to text analytics and demonstrated that no-code Power BI significantly reduced the time and effort required for text analysis tasks compared to traditional coding methods. The finding suggested that no-code tools like Power BI can empower instructors to easily integrate text analytics into their curriculum, fostering student engagement and learning.
[77]	A mixed-methods research design combining questionnaire-based data collection, data mining techniques, and visual analytics	Applied Sciences	The researchers found that visualisations played a crucial role in identifying patterns and trends in the data, which helped in understanding the impact of distinct factors on the research training outcomes. The study also revealed that factors such as the quality of supervision, research resources, and student motivation significantly influenced the success of graduate students in their research endeavours.
[78]	Design involved the development and application of prescriptive analytics, specifically through what-if modelling and machine-learning techniques	Big Data and Cognitive Computing	The study found that using explainable machine learning with automated prescriptive analytics can significantly support students' academic performance. By analysing a large dataset of student information and performance indicators, the system was able to accurately predict students' future academic performance and provide personalised recommendations for interventions to help improve their grades. This approach not only helped identify students who were at risk of falling behind but also provided insights into the underlying factors contributing to their academic struggles. The findings suggest that incorporating advanced analytics tools into educational settings can help educators better support students and improve their academic outcomes.
[79]	A quantitative descriptive design was employed to analyse survey data from 880 Australian Master's students	Studies in Graduate and Postdoctoral Education	The profile of the cohort overall was as expected for an elite academic group, yet there was substantial variation between individuals. Cluster analysis identified three groups of students with meaningfully different dispositional profiles. Exploratory factor analysis revealed two underlying dispositional dimensions, representing epistemic and agentic attributes. Epistemic attributes were most closely related to academic achievement.
[80]	This study employed a mixed-methods approach. Data were collected through a self-report platform and interviews and analysed using machine learning techniques.	Journal of Universal Computer Science	The findings suggest that single-case learning analytics can provide valuable insights for both learners and researchers, contributing to a more personalised and effective learning experience

Author(s)	Design	Publisher	Findings
[81]	A mixed-methods experimental design was used	Education Sciences	The findings demonstrated a positive influence on both academic achievement and student self-regulated learning skills (including metacognitive activities, time management, persistence, and help-seeking behaviours). Additionally, student satisfaction with learning analytics (LA)-based guidance remained high.
[82]	The design involved a quantitative empirical study utilising a questionnaire survey of 155 engineering students	Frontiers in Psychology	The results found that courses on big data analytics have a positive impact on engineering students' abilities in both hard skills and soft skills dimensions, while soft skills have a more significant impact on engineering students' employability.
[83]	The design employed a quantitative approach using online surveys to collect data from 181 postgraduate students in China	Computers in Human Behavior	The study's findings reveal a positive association between mutual trust, social influence, and reward valence within student teams, and their subsequent teamwork engagement. Furthermore, teamwork engagement is shown to be a significant antecedent of learning and work satisfaction among students.
[84]	A mixed-methods research design, integrating a human-centered Artificial Intelligence (AI) framework with experimental validation	Educational Technology and Society	The study achieved a three-pronged advancement: it increased trust in artificial intelligence analysis of learning data, identified key factors influencing learning performance, and determined optimal learning interventions based on artificial intelligence insights.

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